Basics of Data Mining

Data Mining — Week 1: What Do "Data" and "Mining" Actually Mean?

Instructor: Manish Narvariya

Notes by: **Heisenberg** 😎

Index

- Course Logistics — the syllabus fine print, or: how 70% of your fate is decided by one exam.
- Why "Data" Is a Suspiciously Vague Word — UPI receipts, heartbeats, and trees walk into a definition.
- Do We Even Need to Discuss This? — yes, and here's the one-word reason: *solely*.
- Defining Data Properly — mathematical, descriptive, or bust.
- Data as Functions: The Speech Example — your voice, formalized as $x(t)$.
- Why Bother With Functions? Enter the Microphone — where devices quietly lie to you.
- Data as Functions: The Image Example — a tree, reduced to $f(x, y)$.
- Summary: Defining Data — the one-liner worth memorizing.
- What Is "Mining", Really? — averages, transaction counts, and the slow creep toward prophecy.
- The Three Flavors of Mining — query, pattern, and prediction, ranked by how much you should trust them.
- Worked Example: The Laptop Store — one shop, three questions, three very different levels of confidence.
- Key Takeaways — the whole lecture in a nutshell.

Course Logistics

This course, *Basics of Data Mining*, is designed to build fundamentals first rather than jumping straight into applications — the plan is to get the core ideas firmly established and then see how they apply across different domains. Live sessions will be used for feedback on pace, understanding, and expectations, so any confusion should be raised there.

Since the exam is conducted in physical mode, it carries more weight and is considered the more trustworthy and fair component of the evaluation:

Component	Weightage	Notes
Major Exam	70%	Physical mode, main evaluation component
Assignments (×2)	30%	Typically programming assignments, TA-supported, drawn from lecture material

The assignments will be based on material covered in class and the recorded lectures, giving an opportunity to implement concepts and get hands-on practice. The two assignments are intended to be as diverse and interesting as possible.

---

brucks

Why "Data" Is a Suspiciously Vague Word

"Data mining" is a term thrown around so often that most people assume they already understand it. The problem is that unless the *data* and the *mining* are each pinned down precisely, it's impossible to know whether the analysis being attempted is even meaningful — let alone possible — with the data at hand.

So, the natural first question is: **what exactly is data?**

A few examples make the vagueness obvious immediately:

- **A UPI transaction.** Paying ₹100 via UPI creates a record — a timestamp (date and time), the amount, the bank account details, and so on. The moment a transaction happens, a record of *what* happened, *when* it happened, and the *quantities involved* is created. That record is data.
- **ECG data.** An electrocardiogram captures the heartbeat — not exactly periodic, but regular — as a signal derived from vibrations. This is data too, but it comes from a completely different kind of measurement than a UPI transaction.
- **Images, videos, and speech.** A photograph or a video is data. So is speech — fittingly, since this very lecture is speech being recorded. In fact, this single recording contains *three* types of data at once: the speech itself, the text being written, and the video showing it all unfold through time.
- **Weather data.** Rainfall data, satellite images, and similar sources form yet another category, with a meaning completely different from speech, ECG, or transaction data.
- **Hospital records.** A hospital maintains records in a database — patient details, history, symptoms, treatment — which is also data, but this time not necessarily numeric or mathematical in nature.
- Other examples abound: an earthquake signal recorded by a seismograph is data too.

The pattern that emerges is that **data is a vague term** — it can mean different things to different people, in different contexts. A UPI transaction, an ECG trace, a satellite image, and a hospital record are all "data," yet they share almost nothing in common structurally or physically.

Do We Even Need to Discuss This?

This raises a real question, one worth writing down explicitly:

> **Lots of "data" is available, and it is a well-accepted, umbrella term — data mining, machine learning, and AI are all built on it. So do we even need to discuss what "data" means?**

The answer is **yes** — and the reason can be stated in a single underlined word: **solely**.

> All subsequent analysis or mining of the data depends **solely** on how well that data itself is understood.

If the term "data" is left vague, the analysis built on top of it is very likely to be vague, less meaningful, or in some cases outright wrong. This is the entire justification for spending time on something that might otherwise feel like semantics: precision at the data-definition stage is what determines the trustworthiness of everything that follows.

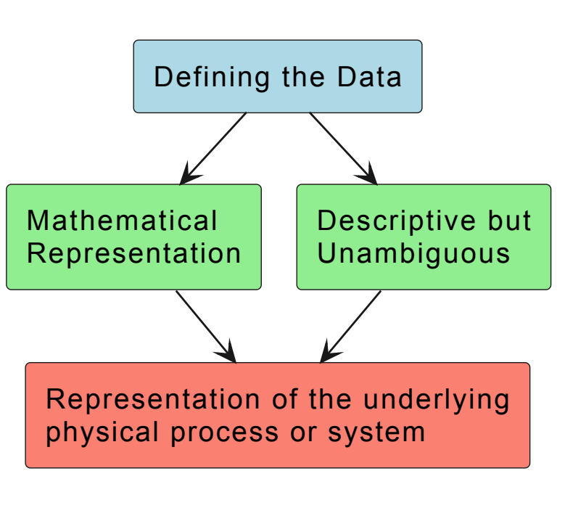
brucks

Defining Data Properly

Defining the data means describing it in one of two ways:

- **Mathematically**, using a precise functional or numerical representation, or
- **Descriptively but unambiguously** — for example, hospital records consist of patient name, date of birth, illness history, prior medication, current symptoms, and ongoing treatment. None of this is mathematical, but it is precise and unambiguous enough that a doctor reading it knows exactly what is meant.

Either way, the representation must capture the **underlying physical process or system** that generated the data.



Defining the Data
Mathematical Representation
Descriptive but Unambiguous
Representation of the underlying physical process or system

This act of **tying data to the system or process that generated it** is a fundamental idea underlying all modeling, data mining, and prediction. Once data is anchored to a physical process, it stops being an abstract blob and starts being something that can be reasoned about rigorously.

Data as Functions: The Speech Example

To formalize this "tying," it helps to return to basic function notation, familiar from school: $f(x)$, where x is the independent variable and $f(x)$ is the dependent value.

For speech (since this very lecture is being spoken and recorded), a new function is defined:

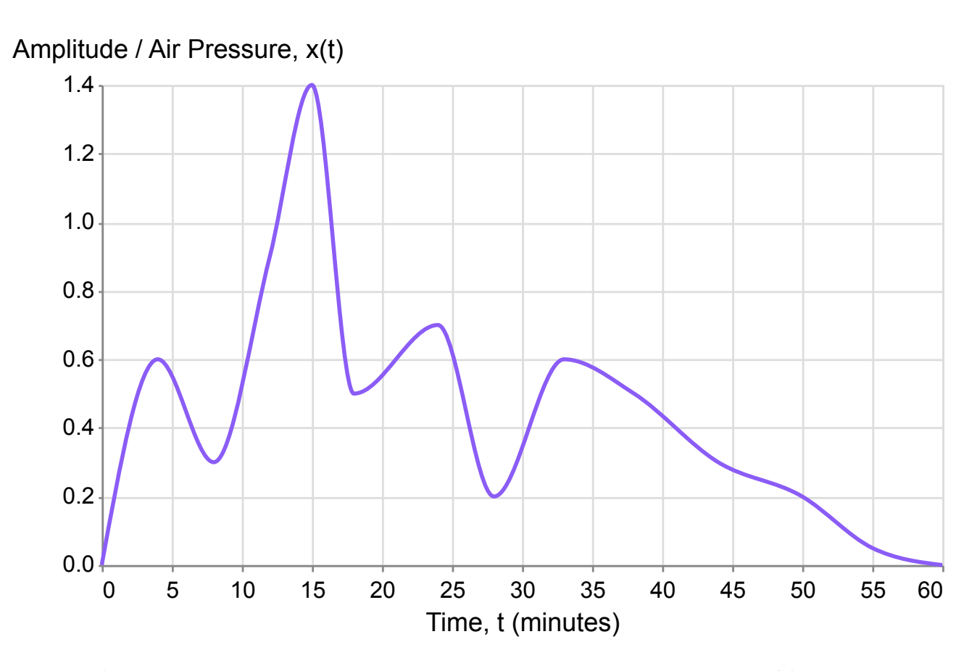
$$x(t)$$

Here t is time, and $x(t)$ represents speech or voice — a function that **maps time to a value**. But *what* value? Physically, speech is a variation of air pressure through the air, produced by vibration of the vocal cords. So:

$$x(t) : \text{time} \rightarrow \text{amplitude of vibration (air pressure)}$$

This is no longer an abstract symbol — both quantities involved are **physically measurable**: time (in minutes, seconds, etc.) and amplitude of vibration (a pressure-related quantity). This is one concrete way of representing data: as a function between physically measurable quantities.

Plotting $x(t)$ against time gives an intuitive picture of a speech signal over the course of a lecture:



At $t = 0$ minutes the lecture begins, and it runs to $t = 60$ minutes. Louder speech (shouting) produces higher amplitude; silence or a pause drives the amplitude toward zero, since there is no air-pressure variation.

> **Note for Review:** The key exam-worthy idea here is not the plot itself, but the *reframing*. Saying "I have speech data" is vague. Saying "I have a function $x(t)$ mapping time to measured air pressure" is precise, physically grounded, and analyzable.

brucks

Why Bother With Functions? Enter the Microphone

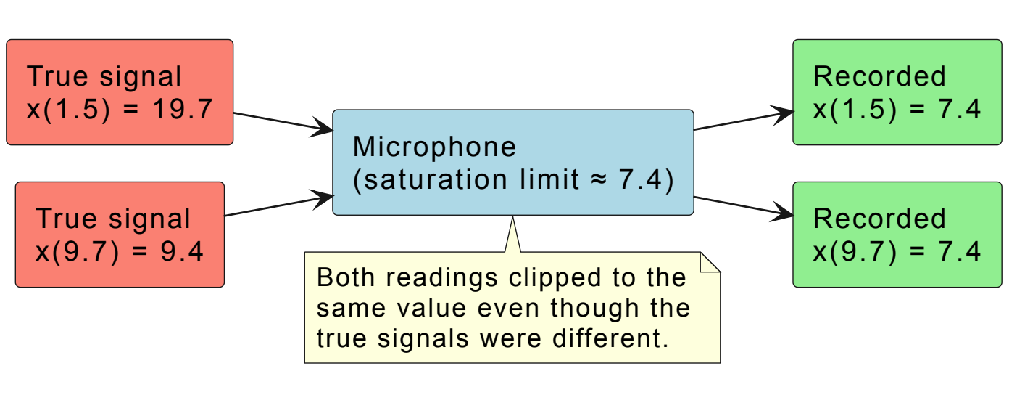
Once data is defined as a function with clear physical quantities — time versus amplitude — a natural follow-up question arises: **why does this framing matter?**

The answer is that defining *what* is being measured immediately points to *how* it is measured — the **device or sensor** that captures the data. For speech, that device is a **microphone**. Every measurement device, digital or analog, has physical limitations: a dynamic range, a saturation point, a sensitivity ceiling. A cheap microphone and an expensive one will not capture the same signal with the same fidelity.

This matters concretely. Suppose the amplitude is measured at two different times:

$$x(1.5) = 7.4 \qquad x(9.7) = 7.4$$

At first glance, these look identical — the same amplitude recorded at two different moments. But if the microphone **saturates** at some maximum level (say, 7.4 in some arbitrary unit), the *true* values could have been very different — perhaps 19.7 at $t = 1.5$ and 9.4 at $t = 9.7$ — with the device simply clipping both readings down to its ceiling of 7.4.



True signal x(1.5) = 19.7
True signal x(9.7) = 9.4
Microphone (saturation limit ≈ 7.4)
Recorded x(1.5) = 7.4
Recorded x(9.7) = 7.4
Both readings clipped to the same value even though the true signals were different.

A downstream machine learning or data mining algorithm has **no way of knowing** that these two "equal" readings of 7.4 are actually artifacts of device saturation rather than genuinely equal signals. Any conclusion drawn from that data risks being less meaningful, or outright incorrect, simply because the physical limitations of the capturing device were ignored.

This is precisely why "I want to analyze speech data" is a weaker statement than "I want to analyze the function that maps time to measured air pressure or amplitude." The second phrasing is more meaningful for two concrete reasons:

1. It states **what physical quantities are actually measured** (time and air pressure), rather than leaving it generic.
2. It allows one to ask **whether the measured signal is accurate** for the intended analysis — i.e., it invites scrutiny of the capturing device's limitations (saturation, dynamic range, distortion, and so on).

Data as Functions: The Image Example

The same idea extends naturally to images. A camera captures a 2D picture, and just as "speech" alone is not a precise enough description, neither is "image." Since an image has two spatial dimensions, it can be defined as a function of two variables:

$$f(x, y)$$

Here x and y are coordinates in space (distances along the horizontal and vertical axes), and $f(x, y)$ is the value of the image at that point — typically its **intensity** (measured, for instance, in candela per square meter).

For a picture of a tree, picking an arbitrary point at coordinates $(34.7, 24.9)$ and measuring its intensity might give:

$$f(34.7, 24.9) = 97.49\ \text{cd/m}^2$$

So $f(x, y)$ maps **distance to intensity**, in the same way $x(t)$ mapped **time to amplitude**. Speech is a **1D signal** (function of a single variable, time); an image is a **2D signal** (function of two spatial variables). Just like the microphone, the camera that produces $f(x, y)$ has its own physical limitations — sensor saturation, dynamic range, and so on — which must be kept in mind for the same reasons discussed above.

brucks

Summary: Defining Data

> **Data should be defined precisely** — not for the sake of rigor alone, but because doing so leads to **more accurate analysis** and a clearer **understanding of the limitations** of that analysis.

Whenever encountering a term like "speech data" or "image data," it is worth pausing to think about the underlying sensor (microphone, camera, or otherwise), whether anything might be missing, whether saturation could have occurred, and whether distortions might exist. These are the questions to ask *before* starting any data mining.

> **Pro-Tip:** A common misconception is that data mining begins with running algorithms on a dataset. In reality, data mining starts *before* that — with understanding how the data was captured and what physical process generated it, so that the limitations of any later analysis are properly understood.

What Is "Mining", Really?

"Mining," like "data," is an umbrella term. In its plain English sense, mining means finding something valuable within a larger body of material — as in mining for gold. Applied to data, the idea is similar, but again it helps to be precise.

Trivial example 1 — Computing an average. Suppose the average marks in an exam are computed. This generates new data — the average — which can be defined mathematically (sum of the numbers, divided by the count):

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

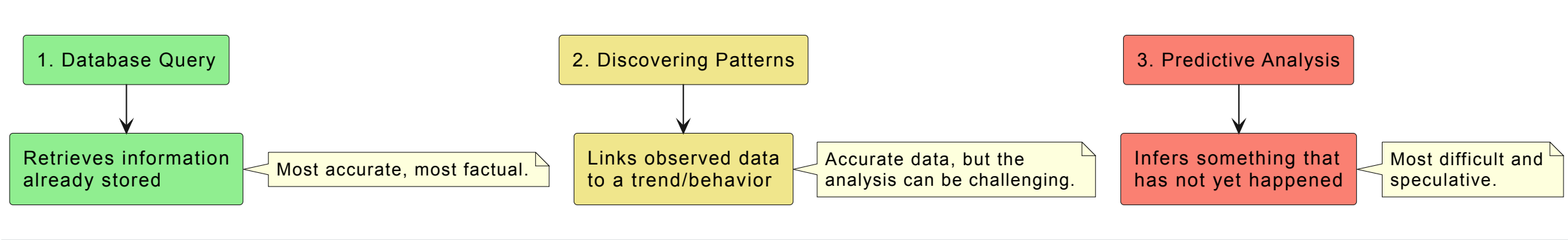
Once this average is obtained, any conclusion drawn *from* it counts as mining. For instance, comparing this batch's average of 63.4 against the previous batch's average of 75.9, and drawing conclusions from that comparison, is a basic — but real — instance of mining.

Trivial example 2 — Counting transactions. A simple question: how many transactions occurred in a store today? Since most transactions are digitized, this can be answered instantly — say, 9800 today. Comparing this against a previous count (e.g., 16000 the next day) and asking *why* it changed is, again, a form of data mining, even though it looks almost trivially simple.

Mining has been practiced, in some form, for centuries — but in the modern **AI era**, the meaning has shifted. It is increasingly associated with **predictive analysis**: not just reporting an average or a transaction count, but looking for patterns across wider datasets — multiple batches, multiple programs, or even nationally — to make a genuine prediction about the future.

The Three Flavors of Mining

Pulling all of this together, "mining" can be understood as spanning three progressively more sophisticated categories:



1. Database Query
Retrieves information already stored
Most accurate, most factual.
2. Discovering Patterns
Links observed data to a trend/behavior
Accurate data, but the analysis can be challenging.
3. Predictive Analysis
Infers something that has not yet happened
Most difficult and speculative.

#	Category	What It Does	Accuracy of Underlying Data	Difficulty of Analysis
1	**Database query**	Retrieves stored information directly (e.g., number of transactions today)	Fully accurate, factual	Trivial — just a lookup
2	**Discovering patterns**	Links observed data to a trend or behavior (e.g., averages rising/falling over years)	Accurate	Challenging — interpretation is non-trivial
3	**Predictive analysis**	Infers something that hasn't happened yet	N/A (forward-looking)	Most difficult, speculative

This three-way split is also echoed in a commonly cited textbook definition of data mining: *"discovering patterns in large amounts of data."* Over time, though, common usage has shifted increasingly toward the third category — predictive analysis.

brucks

Worked Example: The Laptop Store

To make the three categories concrete, consider three questions an inventory manager at a laptop store might ask.

1. "How many laptops were sold last month?" — Deterministic

Since sales records are kept, this can be answered exactly with a single query — say, the answer is a fixed number with **zero uncertainty**. This scenario is called **deterministic**: there is no room for interpretation or misinterpretation. Ask the question, get the number.

2. "Which customers are likely to buy a new laptop in the next 3 months?" — Discovering Patterns

This is one level up in difficulty — it requires predicting **customer behavior**, not just retrieving a count. Based on purchase history, servicing frequency, complaint records, and similar quantifiable factors, a list of likely buyers can be produced. This analysis:

- **Could turn out correct or incorrect** — e.g., out of 10 predicted customers, only 5 might actually buy.
- Even when incorrect, the **underlying reasons were quantifiable** — purchase date, servicing frequency, and complaint counts are all hard facts, even if the *conclusion* drawn from them turns out wrong.

Weather or rainfall prediction falls into this same category: forecasts are built from quantifiable inputs (satellite imagery, wind speed, wind direction), and while they can go wrong, the process itself is scientific and grounded in measurable quantities.

3. "What will the store's turnover be in 2030?" — Predictive Analysis

This is the hardest of the three. Over a horizon of several years, many factors could change — economic shifts, unforeseen events (a drought, a war, a pandemic) — that simply **cannot be quantified in advance**. Any answer here is essentially an informed guess:

- It **might turn out correct or incorrect**, but that isn't the central issue.
- The real issue is that the **underlying reasons are not quantifiable** — unlike the second category, where the reasoning could at least be traced to measurable factors.

Question	Time Horizon	Category	Underlying Reasons Quantifiable?
Laptops sold last month	Instantaneous (past, recorded)	Database query	N/A — fact, not inference
Customers likely to buy in 3 months	Near future	Discovering patterns	Yes
Store turnover in 2030	~4 years out	Predictive analysis	No

A clear trend emerges across these three examples: **as the time horizon extends further into the future, uncertainty increases and the number of quantifiable factors shrinks.** Predictions tend to be more reliable when made "locally" — i.e., over a shorter time frame, or within a well-understood domain or region.

> **Pro-Tip:** A frequent exam trap is treating "correct vs. incorrect" as the deciding factor between categories 2 and 3. It isn't. The real distinction is whether the **reasons behind the analysis are quantifiable**, not whether the final answer happened to be right.

brucks

Key Takeaways

- **Data mining begins with defining data precisely** — mathematically or descriptively — before any algorithm is applied.
- Precise data definitions are always tied to the **underlying physical process or system**, and to the **device or sensor** that captured them (microphone, camera, database, etc.), since every device carries its own limitations (saturation, dynamic range, distortion).
- **Mining** spans a spectrum from simple, fully accurate database queries, through pattern discovery (accurate data, challenging interpretation), to predictive analysis (speculative, most difficult).
- The deciding factor between "acceptable" and "risky" analysis is not whether the final answer is correct, but whether the **reasoning behind it is quantifiable**.
- Wherever possible — and where practical — it is often better to **plan and collect the right data in advance** rather than lean on predictive analysis and its accompanying uncertainty.

The next lecture continues this thread by distinguishing data mining from related terms such as machine learning and statistics, before moving into concrete techniques like clustering and data visualization.